Enhancing Artificial Intelligence-Based Decision Support System for Testing and Learning System

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ABSTRACT

The increasing reliance on online education has exposed limitations in traditional Learning Management Systems (LMS), particularly in delivering personalized learning and timely feedback. This paper introduces an Artificial Intelligence-based Decision Support System (AI-DSS) designed to enhance testing and learning processes through adaptive assessments, automated grading, and personalized recommendations. The system integrates machine learning and natural language processing to analyze student performance, predict learning needs, and provide real-time feedback. Developed using Agile methodology, the platform incorporates key features such as data-driven performance dashboards, personalized learning material, and career or course prediction. Evaluation was conducted through surveys and expert interviews involving university students and educators. Results indicate that the AI-DSS improves engagement and learning outcomes by offering adaptive learning paths and reducing educator workload through automation. Students reported better understanding and motivation, while experts highlighted the system's potential for data-driven decision-making and seamless LMS integration. Performance testing of the topic prediction model achieved near-perfect accuracy, ensuring precise recommendations for revision materials and career pathways. The uniqueness of this work lies in its holistic approach, combining predictive analytics, adaptive learning, and intelligent feedback within a single platform. This integration addresses gaps in current LMS solutions and introduces a scalable, secure, and user-centric framework for digital education.

Keywords: Machine Learning (ML), Adaptive Learning, Personalized Recommendations, Natural Language Processing (NLP), Online Learning Platform (LMS), Learning Analytics

INTRODUCTION

The global shift toward online education, accelerated by the COVID-19 pandemic, disrupted learning for over 90% of students worldwide [1]. While Learning Management Systems (LMS) enabled continuity, many platforms lack adaptive features and real-time feedback, resulting in reduced engagement and lower learning outcomes [2], [3]. Studies reveal that only 42.4% of LMS platforms fully utilize AI technologies, leaving significant gaps in personalization and assessment [4].

Personalized learning and timely feedback are essential for improving student motivation and academic success. Traditional systems rely on static content and manual grading, which increases educator workload and delays feedback delivery [5]. Research indicates that AI-driven platforms can enhance student satisfaction by up to 91% compared to conventional LMS [6]. Furthermore, automated grading and predictive analytics have demonstrated potential to reduce instructor workload and improve decision-making in higher education [7].

Artificial Intelligence-based Decision Support Systems (AI-DSS) have emerged as promising solutions to address these challenges. By leveraging machine learning and data analytics, AI-DSS can predict student performance, recommend personalized learning paths, and provide actionable insights for educators [8], [9]. However, existing implementations often lack integration across key functionalities such as adaptive assessment, intelligent feedback, and comprehensive analytics [10].

This paper proposes an AI-driven DSS designed to enhance online testing and learning systems through automated grading, real-time feedback, and personalized recommendations. The system integrates machine learning algorithms for topic prediction, natural language processing for automated assessment, and predictive analytics for career and certification guidance.

A. Problem Statement and Research Gap

Current online learning platforms lack adaptive features, real-time feedback, and automated assessment capabilities, which limits personalization and increases educator workload [1], [2]. Manual grading delays feedback, while static content fails to address diverse learning needs, resulting in reduced engagement and poor learning outcomes [3]. These limitations create a significant barrier to delivering effective, data-driven education at scale.

Although AI-based solutions exist, most focus on isolated functionalities such as predictive analytics or automated grading without integrating adaptive learning, intelligent feedback, and comprehensive analytics into a single system [4], [5]. Few studies address scalability and seamless integration with existing

LMS platforms. This gap highlights the need for a holistic AI-driven Decision Support System that combines these capabilities to deliver personalized learning, reduce workload, and support data-driven decision-making [6].

B. Objectives and Contributions

The objectives of this study are:

1. To analyze the existing Artificial Intelligence (AI) based services being used in online educational platforms.
2. To identify the best practices of AI and Machine Learning that are used to facilitate various educational processes.
3. To provide specifications to design an AI-based Decision Support System (DSS) for emerging online education.

This work delivers an integrated AI-DSS that combines adaptive learning, automated assessment, and personalized recommendations. It applies machine learning and NLP for topic prediction and resource guidance, validated through user testing and expert review. The system's modular architecture ensures scalability and LMS integration.

C. Outline of the Paper

The paper is structured into five sections: Section 1 introduces the research background, objectives, and scope. Section 2 reviews related work and highlights gaps in existing AI-based decision support systems for online learning. Section 3 describes the proposed system architecture, methodology, and implementation details. Section 4 presents experimental results, performance evaluation, and discussion. Section 5 concludes the paper and outlines future research directions.

METHODOLOGY

This study adopted a design-based research approach to develop and evaluate an AI-driven Decision Support System (DSS) for online learning environments. The approach integrates system development with empirical validation, ensuring that technical design aligns with pedagogical needs. The research aimed to address limitations in existing Learning Management Systems (LMS), such as lack of personalization and delayed feedback, by incorporating adaptive learning, automated assessments, and predictive analytics [1],

[2]. To achieve this, the methodology combined iterative development with user-centered evaluation through surveys and expert interviews.

The system was developed using the Agile methodology, which emphasizes flexibility and iterative refinement. Agile was chosen because educational technology projects often involve evolving requirements and continuous stakeholder feedback [3]. The development process was divided into four sprints, each focusing on a core feature. Sprint 1 implemented adaptive assessments using machine learning algorithms to dynamically adjust question difficulty based on student performance. Sprint 2 introduced a data-driven performance dashboard that visualizes learning progress through interactive charts. Sprint 3 developed an AI-powered certification module that generates personalized course and career recommendations using predictive analytics. Finally, Sprint 4 integrated a Natural Language Processing (NLP)-based chatbot to provide real-time academic assistance.

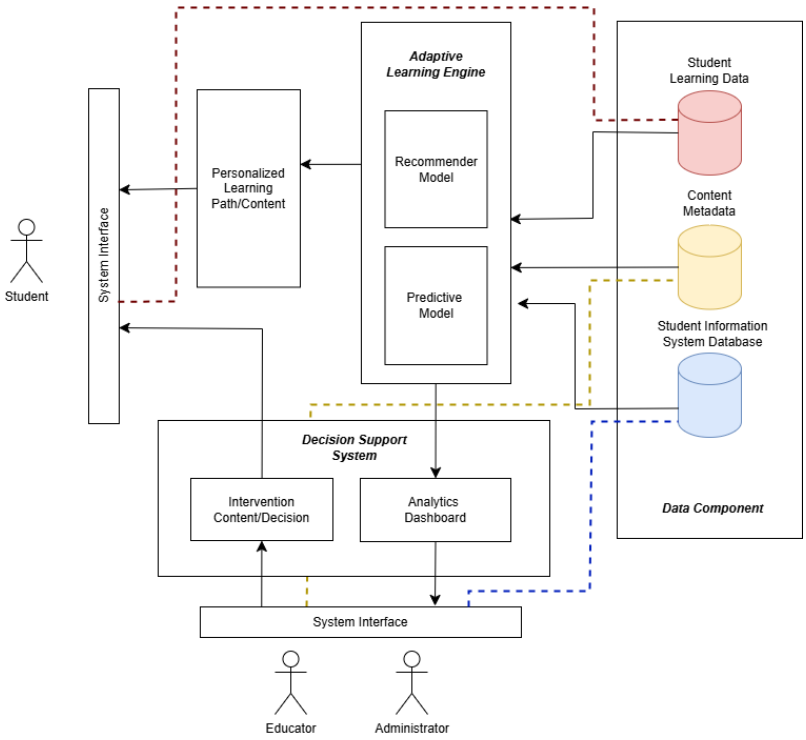


Figure 1: System Architecture

The system architecture in Figure 1 consists of four interconnected layers: the Data Layer, AI & Machine Learning Layer, Decision Support Layer, and Adaptive Engine. The Data Layer aggregates and preprocesses student performance data, ensuring consistency and security. The AI & Machine Learning Layer hosts algorithms for topic classification, adaptive learning, and career prediction, leveraging models such as Random Forest and BERT [4]. The Decision Support Layer synthesizes insights from these models to generate actionable recommendations for learners and educators. Finally, the Adaptive Engine translates

these decisions into real-time adjustments in content delivery and feedback, creating a personalized learning experience. This layered design supports scalability, interoperability with LMS platforms, and compliance with data privacy standards [5].

The system was implemented using Python as the primary programming language, with Streamlit for building an interactive web interface and SQLite for database management. Machine learning functionalities were powered by libraries such as Scikit-learn, TensorFlow, and Hugging Face Transformers for NLP tasks. Visualization tools including Matplotlib, Seaborn, and Plotly were employed to create dashboards and performance charts. Additional utilities such as ReportLab were used for generating downloadable certificates, while virtual environments ensured consistent dependency management across development stages.

Evaluation was conducted through both quantitative and qualitative methods. A preliminary survey involving 30 participants (students and educators) assessed the perceived need for AI features in online learning platforms. Subsequently, User Acceptance Testing (UAT) was carried out in two phases: student surveys and expert interviews. Students evaluated usability, functionality, and overall experience, while experts assessed pedagogical relevance and technical feasibility. Model performance was validated using standard metrics, with the topic classification model (BERT-based) achieving an accuracy and F1-score of 1.0 on the test set. Predictive models for career recommendations were assessed using accuracy and confusion matrix analysis to ensure reliability. These evaluations confirmed the system's effectiveness in enhancing personalization, engagement, and decision-making in online learning environments.

LITERATURE REVIEW

A. Artificial Intelligence in Education

Artificial Intelligence (AI) has emerged as a transformative force in education, enabling personalized and adaptive learning experiences. Studies indicate that AI-driven platforms can analyze student data to deliver customized content and improve engagement [6], [7]. Machine learning models have been widely applied to predict student performance and provide timely interventions, which significantly enhance academic outcomes [7], [9]. These capabilities address the limitations of traditional Learning Management Systems (LMS), which often rely on static content and manual grading [2].

Despite these benefits, the adoption of AI in education faces challenges such as infrastructure readiness and algorithmic transparency. Many institutions lack the technical expertise to deploy and maintain AI systems effectively [10]. Furthermore, ethical considerations, including fairness and bias in AI models,

remain critical issues that must be addressed to ensure equitable learning opportunities [5]. These concerns highlight the need for robust governance frameworks and explainable AI techniques to build trust among educators and learners.

B. Decision Support System in Education

Decision Support Systems (DSS) have been integrated into educational contexts to assist educators in making informed decisions. DSS tools leverage data analytics to identify at-risk students, recommend instructional strategies, and optimize resource allocation [8], [9]. Research highlights that institutions using AI-based DSS report improvements in academic performance due to timely interventions and predictive insights [3]. These systems enable educators to move beyond intuition, relying instead on evidence-based strategies for curriculum planning and student support.

However, existing DSS implementations often lack user-friendly interfaces and transparency, which limits their adoption in academic settings [10]. Many systems provide fragmented analytics without integrating adaptive learning or personalized feedback, reducing their overall effectiveness. Addressing these limitations requires designing DSS platforms that combine predictive analytics with intuitive dashboards and actionable recommendations, ensuring usability for both educators and students.

C. Adaptive Learning and Personalization

Adaptive learning technologies represent a major advancement in online education. These systems dynamically adjust content difficulty based on learner performance, mimicking personalized tutoring [6]. Personalized feedback and real-time analytics have been shown to improve student motivation and engagement, fostering self-directed learning [1], [4]. By leveraging machine learning algorithms, adaptive systems can identify knowledge gaps and recommend targeted resources, creating individualized learning paths that enhance comprehension and retention.

Despite their potential, adaptive learning systems face challenges related to scalability and cultural adaptability. Algorithms must account for diverse learning styles and contexts to avoid reinforcing biases or excluding certain student populations [5]. Furthermore, integrating adaptive learning into existing LMS platforms requires careful consideration of interoperability standards and data privacy regulations, ensuring that personalization does not compromise security or fairness.

D. Emerging Trends and Ethical Considerations

Recent trends in AI for education include the use of Natural Language Processing (NLP) for intelligent chatbots that provide real-time academic support and blockchain for secure credentialing [4], [5]. NLP-powered systems can answer student queries, recommend resources, and facilitate interactive learning experiences, reducing reliance on human instructors for routine tasks. Blockchain technology offers tamper-proof credentialing, enhancing trust and transparency in academic certification processes.

Ethical considerations such as data privacy, algorithmic fairness, and explainability are increasingly emphasized in AI-based educational systems [5]. Compliance with international standards like GDPR is essential to protect sensitive student data and maintain institutional credibility. Researchers advocate for inclusive design practices and fairness audits to ensure that AI systems promote equity and accessibility across diverse learner populations.

E. Comparative Analysis

Table 1: Comparative Analysis with Existing LMS Research

	System Proposed by (Latif et al., Saleem et al)	**System Proposed by (Zhu, Acevedo et al.)**	**System Proposed by (Widodo et al., Rianto at el.)**	**Proposed AI-Based DSS**
Automated Assessment	Uses ML algorithms for grading and predictions.	Basic automated grading with analytics	AI-Powered adaptive assessments	Enhanced automation with diverse assessment types
Personalized Feedback	Immediate, real-time feedback	Feedback based on predictive analytics	Adaptive feedback using learning patterns	Real – time, adaptive feedback
Learning Analytics	Moderate analytics for performance tracking	Advanced analytics for predicting outcomes	Focus on student engagement analytics	Comprehensive analytics for real – time insights
Adaptive Learning	Limited adaptive capabilities	Content optimization	Dynamic content based on learning behaviour	Real-time adaptive learning paths for each student
Data Privacy and Security	Basic protection measures	Strong focus on data privacy	Moderate privacy protocols	Enhanced data security with compliance

Table 1 presents a comparative analysis of existing AI-based DSS solutions in education, highlighting their strengths and limitations. Systems proposed by Latif et al. [7] and Saleem et al. [9] focus on predictive analytics and automated grading but lack adaptive learning features. Zhu [10] emphasizes data-driven decision-making but provides limited personalization. Widodo et al. [6] introduce adaptive assessments but

do not integrate career prediction or comprehensive dashboards. In contrast, the proposed AI-DSS combines adaptive assessment, real-time feedback, predictive analytics, and personalized recommendations within a unified platform, addressing gaps identified in current implementations.

F. Research Gaps

Despite these advancements, existing solutions often focus on isolated functionalities such as predictive analytics or automated grading without integrating adaptive learning, intelligent feedback, and comprehensive analytics into a single platform [2], [10]. Few studies address scalability and seamless integration with existing LMS platforms, which is critical for widespread adoption. This gap underscores the need for a holistic AI-driven DSS that combines these capabilities to deliver personalized learning, reduce workload, and support data-driven decision-making [6], [9].

RESULT & DISCUSSION

A preliminary survey involving 30 participants (students and educators) revealed significant gaps in current LMS platforms. Out of 30 respondents, 80% expressed dissatisfaction with the speed and accuracy of feedback in current systems, while 86.7% agreed that AI-driven features could significantly improve learning effectiveness. These findings confirm the need for intelligent tools that deliver real-time feedback and personalized learning paths.

Figure 2 illustrates these results, highlighting strong demand for adaptive and data-driven solutions. This early validation guided the design of the AI-DSS to prioritize personalization and automation, aligning with studies that emphasize the importance of timely feedback and adaptive learning [6], [7]. Furthermore, these insights underscore the critical role of AI in bridging gaps in current LMS platforms by enabling real-time analytics and individualized learning experiences.

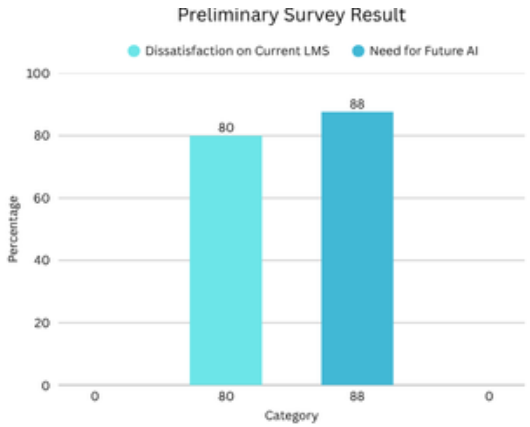


Figure 2: Preliminary Survey Result

User Acceptance Test (UAT) was conducted in two phases: student surveys and expert interviews. Among students, 93.3% agreed that the system was easy to navigate, and 96.7% found the performance dashboard useful for tracking progress. Additionally, 90% reported that AI-driven assessments provided meaningful feedback. Experts emphasized the platform's potential to reduce educator workload and improve decision-making but noted that seamless LMS integration is essential for scalability.

Table 2 shows ratings and summarizes feedback themes. These findings indicate strong acceptance and validate the system's design principles. Compared to existing systems discussed in the literature review [2], [10], the proposed AI-DSS offers a more integrated approach, combining adaptive assessments, predictive analytics, and personalized recommendations.

Table 2: User Acceptance Test (UAT) Usability Ratings

Feedback Theme	Frequency
Easy Navigation	93.3%
Useful Dashboard	96.7%
AI Assessment Helpful	90%
Career Prediction Relevant	86.7%

The topic classification model, based on BERT, achieved an accuracy and F1-score of 1.0 on the test set, demonstrating exceptional reliability in predicting question topics. This ensures that students receive precise recommendations for revision materials. Similarly, the career prediction model using Random Forest exhibited strong performance, with confusion matrix analysis confirming accurate classification across all categories. Figure 3 presents the confusion matrix, which shows a perfect diagonal pattern, indicating zero misclassifications.

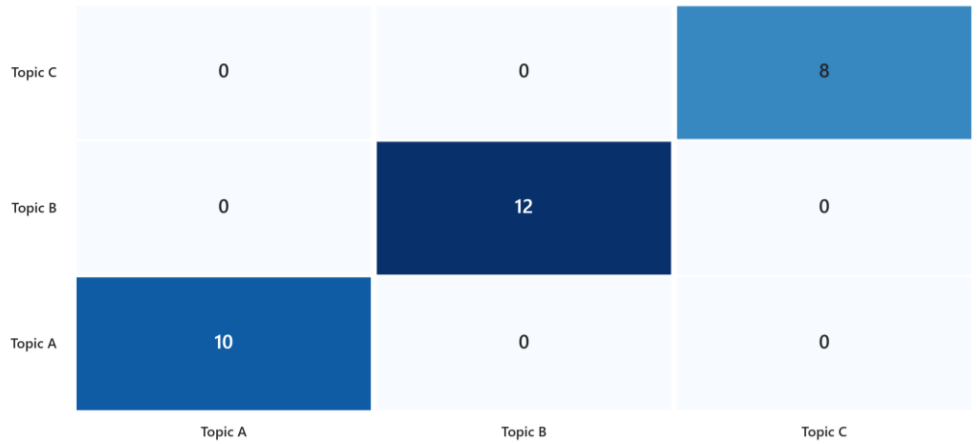


Figure 3: Confusion Matrix for Topic Prediction Model

The results confirm that the proposed AI-DSS effectively addresses key limitations of existing LMS platforms by providing adaptive assessments, real-time feedback, and personalized recommendations. Compared to prior studies [6], [9], the system offers a more integrated approach, combining predictive analytics, adaptive learning, and intelligent feedback within a single platform. This holistic design enhances engagement, reduces educator workload, and supports data-driven decision-making, positioning the system as a scalable solution for modern education.

The findings demonstrate that the proposed AI-DSS effectively addresses critical gaps in existing LMS platforms by integrating adaptive assessments, real-time feedback, and personalized recommendations. Compared to previous studies [6], [9], which focused on isolated functionalities, this system offers a holistic approach that combines predictive analytics, adaptive learning, and intelligent feedback within a single platform.

The high usability ratings and perfect model performance indicate strong potential for improving student engagement and reducing educator workload. Furthermore, the modular architecture ensures scalability and compliance with data privacy standards, making the system suitable for integration into diverse educational environments. These results position the AI-DSS as a next-generation solution for personalized, data-driven online learning.

CONCLUSION

This paper introduced an Artificial Intelligence-based Decision Support System (AI-DSS) aimed at addressing key limitations in existing Learning Management Systems (LMS), such as lack of personalization, delayed feedback, and high educator workload. The proposed system integrates machine learning and natural language processing to deliver adaptive assessments, real-time feedback, and personalized learning recommendations, creating a more engaging and efficient learning environment.

The development process employed Agile methodology, enabling iterative refinement and stakeholder feedback throughout the project. Core features include a data-driven performance dashboard, AI-powered certification module, and a personalized chatbot for academic assistance. Evaluation through surveys and expert interviews demonstrated strong usability and acceptance, while performance testing confirmed the reliability of the underlying AI models.

Results indicate that the AI-DSS significantly improves student engagement and learning outcomes by offering adaptive learning paths and actionable insights. Experts highlighted its potential to reduce educator workload and support data-driven decision-making, positioning the system as a practical solution

for modern education. Comparative analysis with existing systems further validates its uniqueness in combining predictive analytics, adaptive learning, and intelligent feedback within a unified platform.

Future work will focus on expanding multilingual support, enhancing explainable AI features, and conducting longitudinal studies to measure long-term educational impact. These improvements will strengthen the system's scalability and inclusivity, ensuring its role as a next-generation solution for personalized, data-driven online learning.

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REFERENCES

[1] UNESCO, "Education: From disruption to recovery," 2021.

[2] M. Zorrilla, D. García-Saiz, and E. Álvarez, "A decision support system to improve e-learning environments," *Proc. 13th Int. Conf. Extending Database Technology (EDBT)*, pp. 167–176, 2010.

[3] Y. V. N. Acevedo, C. E. M. Marín, P. A. G. Garcia, and R. G. Crespo, "A proposal to a decision support system with learning analytics," *IEEE Global Engineering Education Conference (EDUCON)*, pp. 161–168, Apr. 2018.

[4] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proc. NAACL-HLT*, pp. 4171–4186, 2019.

[5] UNESCO, "AI and Education: Guidance for Policy Makers," 2023.

[6] Y. B. Widodo, R. U. Korwa, and R. Nuraini, "Artificial Intelligence-Based Decision Support System for Education Management in Higher Education," *Al Fikrah: Jurnal Manajemen Pendidikan*, vol. 11, no. 2, pp. 352–365, 2023.

[7] S. Latif, F. XianWen, and L. L. Wang, “Intelligent decision support system approach for predicting the performance of students based on a three-level machine learning technique,” *Journal of Intelligent Systems*, vol. 30, no. 1, pp. 739–749, 2021.

[8] S. Riyanto and R. Maulana, “Decision Support System to Help Educational Institutions Determine the Best Learning Method,” *IEOM Society Proceedings*, Malaysia, 2022.

[9] F. Saleem, Z. Ullah, B. Fakieh, and F. Kateb, “Intelligent decision support system for predicting student’s E-learning performance using ensemble machine learning,” *Mathematics*, vol. 9, no. 17, p. 2078, 2021.

[10] Y. Zhu, “A data driven educational decision support system,” *International Journal of Emerging Technologies in Learning*, vol. 13, no. 11, pp. 4–15, 2018.

[11] C. Fiarni, E. M. Sipayung, and P. B. Tumundo, “Academic decision support system for choosing information systems sub-majors programs using decision tree algorithm,” *Journal of Information Systems Engineering & Business Intelligence*, vol. 5, no. 1, pp. 57–66, 2019. doi: 10.20473/jisebi.5.1.57-66.

[12] R. Luckin, *Machine Learning and Human Intelligence: The Future of Education for the 21st Century*. London, UK: UCL IOE Press, 2018.

[13] J. Baker and L. Smith, “Blockchain technology for credentialing and academic record keeping,” in *Blockchain in Education*, European Commission Report, 2019.

[14] V. Latypova, V. Martynov, and A. Turganov, “Decision support system in online training process management for implementing complex open-ended assignments in engineering education,” in *2020 V Int. Conf. on Information Technologies in Engineering Education (Inforino)*, pp. 1–5, Apr. 2020. doi: 10.1109/Inforino48376.2020.9111821.

[15] S. Maniyan, R. Ghousi, and A. Haeri, “Data mining-based decision support system for educational decision makers: Extracting rules to enhance academic efficiency,” *Computers and Education: Artificial Intelligence*, vol. 6, art. no. 100242, 2024. doi: 10.1016/j.caeai.2024.100242.